1.1 Right-angle triangle with numbers:

```
// Name: Tanushree Lohiya
// Roll no: SEA56
// Right-angle triangle with
numbers
#include<stdio.h>
void main(){
   int n = 4;
   for(int i = 1; i <= n; i++){
      for(int j = 1; j <= i; j++){
         printf ("%d ", j);
      }
      printf("\n");
   }
}
```

Output:

```
1
1 2
1 2 3
1 2 3 4
```

1.2 Diamond shape with numbers:

```
// Name: Tanushree Lohiya
// Roll no: SEA56
// Diamond shape with
numbers
#include<stdio.h>
void main(){
   int n = 4;
   for(int i = 1; i <= n; i++){
      for(int j = n-1; j >= i; j--){
         printf(" ");
      }
      for(int j = i; j > 0; j--){
         printf("%d ", i);
      }
      printf("\n");
   }
   for(int i = n-1; i > 0; i--){
      for(int j = i; j < n; j++){
         printf(" ");
      }
      for(int j = 0; j < i; j++){
         printf("%d ", i);
      }
      printf("\n");
   }
}
```

Output:

```
   1
  2 2
 3 3 3
4 4 4 4
 3 3 3
  2 2
   1
```

1.3 Pyramid with asterisks:

```
// Name: Tanushree Lohiya  // Roll no: SEA56
// Pyramid with asterisks
#include<stdio.h>
void main(){
   int n = 4;
   for(int i = 1; i <= n; i++){
      for(int j = n-1; j >= i; j--){
         printf(" ");
      }
      for(int j = i; j > 0; j--){
         printf("* ");
      }
      printf("\n");
   }
}
```

Output:

```
   *
  * *
 * * *
* * * *
```

1.4 Pyramid using alphabets:

```
// Name: Tanushree Lohiy
a
// Roll no: SEA56
// Pyramid using
alphabets
#include<stdio.h>
void main(){
   int n = 4;
   for(int i = 0; i < n; i++){
      char ch = 'A';
      for(int j = n-1; j > i; j--){
         printf(" ");
      }
      for(int j = 0; j <= i; j++){
         printf("%c", ch+j);
      }
      for(int j = i; j > 0; j--){
         printf("%c", ch+j-1);
      }
      printf("\n");
}
```

Output:

```
    A
   ABA
  ABCBA
ABCDCBA
```

2. Student Result Database Management using Array of Structures

```
// Name: Tanushree Lohiya
// Roll no: SEA56
// Student Result Database Management using Array of Structures

#include<stdio.h>

typedef struct stud{
   int rn;
   char name[20];
   char program[20];
   char sub1[10], sub2[10];
   int sub1_mk, sub2_mk, total;
   float avg;
}stud;

void main(){
   stud s[20];
   int n, op, key;
   void create(stud[], int);
   void display(stud[], int);
   void search(stud[], int, int);
   printf("STUDENTS DATABASE MANAGEMENT\n");

   do{
      printf("\n1. Create\n2. Display\n3. Search\n");
      printf("Enter option no. : ");
      scanf("%d", &op);
      switch(op){
         case 1:
         printf("\nEnter no. of students: ");
         scanf("%d", &n);
         create(s, n);
         break;
         case 2:
         display(s, n);
         break;
         case 3:
         search(s, n, key);
         break;
      }
   }while(op < 3);
```

```
        }
void create(stud s[], int n){
   printf("\n---//ENTER STUDENTS DATA//---\n");
   for(int i = 1; i <= n; i++){
      printf("\n~~~Student %d Data~~~\n", i);
      printf("Enter student's name: ");
      scanf("%s", &s[i].name);
      printf("Enter student's roll no. : ");
      scanf("%d", &s[i].rn);
      printf("Enter student's program: ");
      scanf("%s", &s[i].program);
      printf("Enter subject1: ");
      scanf("%s", &s[i].sub1);
      printf("Enter subject2: ");
      scanf("%s", &s[i].sub2);
      printf("Enter subject 1's marks: ");
      scanf("%d", &s[i].sub1_mk);
      printf("Enter subject 2's marks: ");
      scanf("%d", &s[i].sub2_mk);


            s[i].total = s[i].sub1_mk + s[i].sub2_mk;
            s[i].avg = s[i].total/2;
         }
      }

      void display(stud s[], int n){
         printf("\n---//DISPLAYING STUDENTS DATA//---");
         for(int i = 1; i <= n; i++){
            printf("\n\n~~~Student %d Data~~~", i);
            printf("\nName: %s", s[i].name);
            printf("\nRoll no. : %d", s[i].rn);
            printf("\nProgram: %s", s[i].program);
            printf("\nSubject 1: %s", s[i].sub1);
            printf("\nSubject 2: %s", s[i].sub2);
            printf("\nSubject 1's marks: %d", s[i].sub1_mk);
            printf("\nSubject 2's marks: %d", s[i].sub2_mk);
            printf("\nTotal: %d", s[i].total);
            printf("\nAverage: %.2lf", s[i].avg);
         }
         printf("\n\n---//END//---\n");
      }

      void search(stud s[], int n, int key){
```

```
    int flag = 0;
    printf("\nEnter student's roll no. to search: ");
    scanf("%d", &key);
    for(int i = 1; i <= n; i++){
        if(s[i].rn == key){
            printf("STUDENT FOUND\n");
            flag++;
            break;
        }
    }
    if(flag == 0){
        printf("STUDENT NOT FOUND\n");
    }
}
```

Output:

```
STUDENTS DATABASE MANAGEMENT

1. Create
2. Display
3. Search
Enter option no. : 1

Enter no. of students: 2

---//ENTER STUDENTS DATA//---

~~~Student 1 Data~~~
Enter student's name: Tanushr
ee
Enter student's roll no. : 56
Enter student's program: EnTC
Enter subject1: DSA
Enter subject2: DE
Enter subject 1's marks: 90
Enter subject 2's marks: 95
~~~Student 2 Data~~~
Enter student's name: Tanu
Enter student's roll no. : 56
Enter student's program: EnTC
Enter subject1: DSA
Enter subject2: DE
```

```
Enter subject 1's marks: 98
Enter subject 2's marks: 100

1. Create
2. Display
3. Search
Enter option no. : 2

---//DISPLAYING STUDENTS DATA//---

~~~Student 1 Data~~~
Name: Tanushree
Roll no. : 56
Program: EnTC
Subject 1: EC
Subject 2: DE
Subject 1's marks: 90
Subject 2's marks: 95
Total: 185
Average: 92.00

~~~Student 2 Data~~~
Name: Tanu
Roll no. : 56
Program: EnTC
Subject 1: EC
Subject 2: DE
Subject 1's marks: 98
Subject 2's marks: 100
Total: 198
Average: 99.00

---//END//---

1. Create
2. Display
3. Search
Enter option no. : 3

Enter student's roll no. to search: 15
STUDENT NOT FOUND
```

3. Searching contact name using Sequential and Binary Search

```
// Name: Tanushree Lohiya
// Roll no: SEA56
// Searching contact name using Sequential and Binary
Search
#include <stdio.h>
#include <string.h>

typedef struct contact{
   char name[20];
   char phone[20];
   char address[20];
}contact;

void main(){
   contact c[20];
   int n, i;
   int lincom = 0;
   int linfound = 0;
   int bincom = 0;
   char key[20];

   printf("Enter number of contacts: ");
   scanf("%d", &n);

   printf("\n---//ENTER CONTACT DETAILS (A-Z)//---\n");
   for(i = 1; i <= n; i++){
      printf("\n~~~Contact %d~~~\n", i);
      printf("Name: ");
      scanf("%s", &c[i].name);
      printf("Contact: ");
      scanf("%s", &c[i].phone);
      printf("Address: ");
      scanf("%s", &c[i].address);
   }

   printf("\n---//DISPLAYING CONTACTS//---\n");
   for(i = 1; i <= n; i++){
      printf("\n~~~Contact %d~~~", i);
      printf("\nName: %s", c[i].name);
      printf("\nContact: %s", c[i].phone);
      printf("\nAddress: %s", c[i].address);
```

```
}
printf("\n\n---//END//---\n\n");

printf("Enter name of contact to search: ");
scanf("%s", &key);

printf("\n---//Linear Search//---");
for(i = 1; i <= n; i++){
   lincom++;
   if(strcasecmp(key, c[i].name)==0){
      printf("\nCONTACT FOUND");
      printf("\nName: %s", c[i].name);
      printf("\nContact: %s ", c[i].phone);
      printf("\nAddress: %s", c[i].address);
      linfound = 1;
      break;
   }
}
if(linfound == 0){
   printf("\nCONTACT NOT FOUND");
}

printf("\nLinear comparisons: %d", lincom);
printf("\n\n---//Binary Search//---");

int high = n;
int low = 1;
int mid;

while(low <= high){
   mid = (high+low)/2;
   if(strcasecmp(c[mid].name, key) == 0){
      bincom++;
      printf("\nContact found at location %d", mid);
      break;
   }else if(strcasecmp(c[mid].name, key)<0){
      bincom++;
      low = mid+1;
   }else{
      bincom++;
      high = mid-1;
   }
}
if(low>high){
```

```
        printf("\nContact not found");
    }

    printf("\nNumber of binary comparisons: %d", bincom);

}
```

Output:

```
Enter number of contacts: 5

---//ENTER CONTACT DETAILS //---

~~~Contact 1~~~
Name: Tanushree
Contact: 123
Address: Amravati

~~~Contact 2~~~
Name: Tanu
Contact: 456
Address: Akola

~~~Contact 3~~~
Name: Tanuja
Contact: 789
Address:
Sangamner
~~~Contact 4~~~
Name: Ram
Contact: 901
Address: Nagpur

~~~Contact 5~~~
Name: shyam
Contact: 112
Address: Kolkata

---//DISPLAYING CONTACTS//---

~~~Contact 1~~~
Name: Tanushree
Contact: 123
Address: Amravati
```

~~~Contact 2~~~
Name: tanu
Contact: 456
Address: Akola
~~~Contact 3~~~
Name: Tanuja
Contact: 789
Address: sangam
ner
~~~Contact 4
~~

Name: Ram
Contact: 901
Address: Nagpur
~~~Contact 5~~~
Name: shyam
contact:112
Address: kolkata

---//END//---
Enter name of contact to search: pushpak
---//Linear Search//---
CONTACT FOUND
Name: Tanu
Contact: 456
Address: Akola
Linear comparisons: 4

---//Binary Search//---
Contact found at location 4
Number of binary comparisons: 2

4.Decimal to Binary Conversion using Stack

```
// Name: Tanushree Lohiya
// Roll no: SEA56
// Decimal to Binary Conversion using Stack

#include<stdio.h>
void main(){
   int n;
   printf("Enter Decimal number: ");
   scanf("%d", &n);
   int s[10], top = -1, r;

   while(n != 0){
      r = n%2;
      n = n/2;
      top++;
      s[top] = r;
   }

   printf("Binary number for given decimal number is: ");
   for(int i = top; i >= 0; i--){
      printf("%d", s[i]);
   }
}
```

Output:

```
Enter Decimal number: 15
Binary number for given decimal number is: 1111
```